Section Minimum Duration : 0
Section Time In : Minutes
Maximum Instruction Time : 0
Sub-Section Number : 1
Sub-Section Id : 640653168507
Question Shuffling Allowed : No

Question Number : 66 Question Id : 6406531113758 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406533774138. ✔ YES
6406533774139. ✖ NO

Sub-Section Number : 2
Sub-Section Id : 640653168508
Question Shuffling Allowed : Yes

Question Number : 67 Question Id : 6406531113759 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the following is the correct way of calling an external CSS file using HTML?

Options :

```
<head>
<link url="stylesheet" type="text/css" href="mystyle.css">
</head>
```

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

```
<head>
<link href="stylesheet" type="text/css" file="mystyle.css">
</head>
```

```
<head>
<link url="stylesheet" type="text/css" file="mystyle.css">
</head>
```

Question Number : 68 Question Id : 6406531113760 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which command would you use to get **detailed information** about the HTTP request and response, **including headers**, and to **help debug the HTTP request** when using curl?
Options :

```
curl -i <URL>
```

```
curl -v <URL>
```

```
curl --headers <URL>
```

```
curl -X GET <URL>
```

Question Number : 69 Question Id : 6406531113761 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Read the statements given below carefully and select the correct option.

Statement 1: If an element that belongs to two different classes is styled externally using both the classes, then for the same attribute, it will acquire styling from the latest class in order.
Statement 2: If an element having an ID and a class is styled externally using both its ID and the class, then for the same attribute, it will acquire styling from the ID selector.
Options :

Sub-Section Number :	3
Sub-Section Id :	640653168509
Question Shuffling Allowed :	Yes

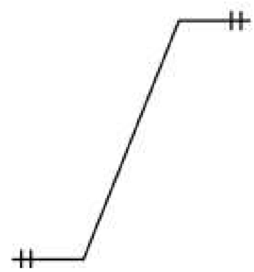
Question Number : 70 Question Id : 6406531113762 Question Type : MSQ

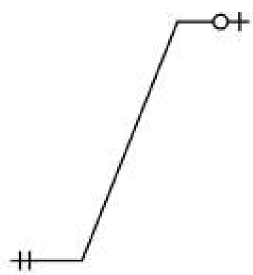
Correct Marks : 2 Max. Selectable Options : 0

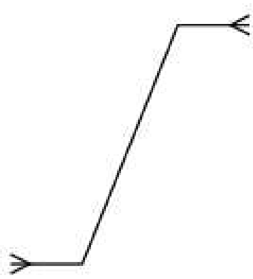
Question Label : Multiple Select Question

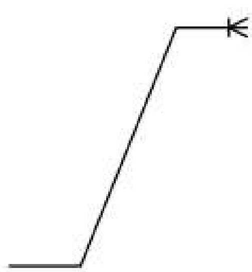
Which of the following options represents a one-to-one relationship in the Entity-Relationship Diagram?

Options :









Sub-Section Number :	4
Sub-Section Id :	640653168510
Question Shuffling Allowed :	Yes

Question Number : 71 Question Id : 6406531113763 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following HTML document:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <style>
        #container {
            border-style: dashed;
            background-color: red;
        }
        .content {
            border-style: solid;
            border-color: green;
            background-color: orange;
        }
        div {
            width: 250px;
            border: 3px dotted black;
            border-color: yellow;
        }
    </style>
</head>
<body>
    <div id="container" class="content" style="background-color: yellow;">
        <h2>Welcome to Web Dev</h2>
    </div>
</body>
</html>
```

How will the browser render the above HTML document?

Options :

Question Number : 72 Question Id : 6406531113764 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following HTML code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
   <title>Document</title>
   <style>
       div {
           color: blue;
       }
       #s1 {
           color: yellow;
       }
       .s2{
           color: green;
       }
       h3{
           color: red;
       }
   </style>
</head>
<body>
   <div class="s2" id="s1"><h3>IITM Online BS Degree Program</h3></div>
</body>
</html>
```

What is the colour of the printed text “IITM Online BS Degree Program”?

Options :

BLUE

YELLOW

GREEN

RED

Question Number : 73 Question Id : 6406531113765 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the template code and Python code given below:

Template code: “output.html”

```
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
   <ol>
       {%for p in products%}
           {%if "i" in p.name|string %}
               <li>{{p.id}}, {{p.name}}, {{p.qty}}, {{p.price}}</li>
           {%endif%}
       {%endfor%}
   </ol>
</body>
</html>
```

Python code: “main.py”

```
from jinja2 import Template

products = [
   {"id": "101", "name": "Food", "qty": 78, "price": 100},
   {"id": "102", "name": "Electrical", "qty": 4, "price": 16},
   {"id": "103", "name": "Stationary", "qty": 7, "price": 14},
   {"id": "104", "name": "Electronics", "qty": 10, "price": 145},
]

with open("output.html") as r:
   t = Template(r.read())
   print(t.render(products=products))
```

What will the output of the above Python code be?

Options :

```
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
    <ol>
        <li>101, Food, 78, 100</li>
        <li>103, Stationary, 7, 14</li>
    </ol>
</body>
</html>
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
    <ol>
        <li>102, Electrical, 4, 16</li>
        <li>103, Stationary, 7, 14</li>
        <li>104, Electronics, 10, 145</li>
    </ol>
</body>
</html>
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
    <ol>
        <li>103, Stationary, 7, 14</li>
    </ol>
</body>
</html>
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
    <ol>
        <li>102, Electrical, 4, 16</li>
        <li>104, Electronics, 10, 145</li>
    </ol>
</body>
</html>
```

Question Number : 74 Question Id : 6406531113766 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following Python code snippet:

Python code: “main.py”

```
import sys
movie_ratings = {
   "Kalki": [5, 1, 2, 1, 1],
   "Jailer": [7, 3, 1, 2, 1],
   "Leo": [4, 3, 2, 2, 1],
}

def do_something(movie_name):
   avg_ratings = {}
   for movie in movie_ratings:
       if movie == movie_name:
           avg_ratings[movie] = sum(movie_ratings[movie]) /
len(movie_ratings[movie])
           return avg_ratings
   return "Movie not found"

args = sys.argv
print(do_something(args[1]))
```

The above file will result in the output as ***“2.8”***, for which of the following command line input?

Options :

python main.py Kalki
python main.py Leo
python main.py Pushpa
python main.py Jailer

Question Number : 75 Question Id : 6406531113767 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Consider two python files, mad1.py and mad2.py with following code snippets.

File1: mad1.py

```
import sys
import mad2
print('Welcome' + ' ' + sys.argv[1])
```

File2: mad2.py

```
import sys
print('Welcome' + ' ' + sys.argv[0])
```

What is the output of the following command “python mad1.py mad2.py MAD-I MAD-II”?

Options :

Welcome MAD-I.py
Welcome MAD-II.py

Welcome mad2.py
Welcome mad1.py

Welcome mad1.py
Welcome mad2.py

Welcome MAD-II.py
Welcome MAD-I.py

Sub-Section Number :	5
Sub-Section Id :	640653168511
Question Shuffling Allowed :	Yes

Question Number : 76 Question Id : 6406531113768 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Based on the provided latency and response size information, which of the following statements is/are correct?

(Assume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

Options :

The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Question Number : 77 Question Id : 6406531113769 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the following two tables in an SQLite database:

Table: Gifts

GiftID	GiftName	Price
1	Teddy Bear	250
2	Coffee Mug	150
3	Keychain	100
4	Notebook	180
5	Wall Clock	300

Table: Reviews

ReviewID	GiftID	Rating
1	1	4.2
2	2	4.5
3	3	3.9
4	4	4.7
5	5	4.0

Which of the following SQL queries correctly retrieves a list of **unique gift names** with a **price less than 200** rupees and a **rating above 4**? (**Note:** Displaying the rating in the result is optional, but duplicate gift names should not appear.)

Options :

```
SELECT GiftName, Price
FROM Gifts,Reviews
WHERE Price < 200 AND Rating > 4;
```

```
SELECT g.GiftName, g.Price
FROM Gifts g
JOIN Reviews r ON g.GiftID = r.GiftID
WHERE g.Price < 200 AND r.Rating > 4;
```

```
SELECT g.GiftName, g.Price, r.Rating
FROM Gifts, Reviews
WHERE g.Price < 200
AND r.Rating > 4;
```

```sql
SELECT g.GiftName, g.Price, r.Rating
FROM Gifts g, Reviews r
WHERE g.GiftID = r.GiftID
AND g.Price < 200
AND r.Rating > 4;
```

Sub-Section Number :	6
Sub-Section Id :	640653168512
Question Shuffling Allowed :	Yes

Question Number : 78 Question Id : 6406531113770 Question Type : MCQ
Correct Marks : 4.5

Question Label : Multiple Choice Question

A server S requests to retrieve data from two data centers DC_1 and DC_2, which are located at distances of D_1 and D_2 meters respectively from the server S. The speed of data travelling through the medium that connects S with DC_1 is m_1 m/sec and that connects S with DC_2 is m_2 m/sec. With the given media for data transfer between S $\rightarrow DC_1$ and S $\rightarrow DC_2$, it takes an additional 200 milliseconds for the data to reach S from DC_2 than it takes from DC_1 if the complete response is ready only when the data is available to S from both the datacenters. Figure out the value of m_2 (in m/sec) in terms of D_1, D_2 and m_1. [Note: The data centers send responses only when they receive a request from the server]

Options :

$$m_2 = \frac{D_2 m_1}{D_1 - 0.2m_1}$$

$$m_2 = \frac{D_2 m_1}{D_1 + 0.2m_1}$$

$$m_2 = \frac{2D_2 m_1}{2D_1 - 0.2m_1}$$

$$m_2 = \frac{2D_2 m_1}{2D_1 + 0.2m_1}$$

Question Number : 79 Question Id : 6406531113771 Question Type : MCQ
Correct Marks : 4.5
Question Label : Multiple Choice Question

Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen **(-)**.
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon **(:)**
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

Options :

`0000:3B2B:9821:0652`

`3B2B:9821:0652:0000`

`0000:0652:9821:3B2B`

`0652:9821:3B2B:0000`

Question Number : 80 Question Id : 6406531113772 Question Type : MCQ
Correct Marks : 4.5
Question Label : Multiple Choice Question

Consider the following code:

```
import pyhtml as html

table = {'table': [[1,2,3],[4,5,6]]}

def f_table(ctx):
  template_str = html.html(
    html.head(
      html.title('Title')
    ),
    html.body(
      html.table(
        (html.tr(
            [html.td(cell) for cell in row]
          ) for row in ctx['table'])
      )
    )
  )
  return (template_str)

template = f_table(table)
print(template.render())
```

What will the output of the above PyHTML code be?

Options :

```
<!DOCTYPE html>
<html>
  <head>
    <title>Title</title>
  </head>
  <body>
    <table>
      <tr>
        <td>1</td>
        <td>2</td>
        <td>3</td>
      </tr>
      <tr>
        <td>4</td>
        <td>5</td>
        <td>6</td>
      </tr>
    </table>
  </body>
</html>
```

```
<!DOCTYPE html>
<html>
  <head>
    <title>Title</title>
  </head>
  <body>
    <table>
      <tr>
        <td>1</td>
        <td>2</td>
        <td>3</td>
        <td>4</td>
        <td>5</td>
        <td>6</td>
      </tr>
    </table>
  </body>
</html>
```

```
AttributeError: module 'pyhtml' has no attribute 'html'.
```

None of these

Sub-Section Number : 7
Sub-Section Id : 640653168513
Question Shuffling Allowed : No

Question Id : 6406531113773 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (81 to 82)

Question Label : Comprehension

app.py

```
import sys
from string import Template as T1
from jinja2 import Template as T2

temp = "The book title is $title and it costs {{price}}."
if sys.argv[1] == 'jinja 2':
    temp = T1(temp)
    out = temp.substitute({'title': 'Python Basics', 'price': '$20'})
elif sys.argv[1] == 'string':
    temp = T2(temp)
    out = temp.render({'title': 'Python Basics', 'price': '$20'})
else:
    out = 'Invalid input'
print(out)
```

Based on the above program, answer the given subquestions.
Sub questions

Question Number : 81 Question Id : 6406531113774 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
What will be the output of the given Python code snippet on the terminal for the command python app.py string ?

Options :

```
The book title is Python Basics
and it costs $20.
```

```
The book title is $title and it
costs $20.
```

```
The book title is Python Basics
and it costs {{price}}.
```

```
The book title is $title and it
costs {{price}}.
```

Question Number : 82 Question Id : 6406531113775 Question Type : MCQ
Correct Marks : 4.5
Question Label : Multiple Choice Question
What will be the output of the given Python code snippet on the terminal for the command python app.py jinja 2 ?

Options :

```
The book title is Python Basics and
it costs $20.
```

```
The book title is $title and it
costs $20.
```

```
The book title is Python Basics and
it costs {{price}}.
```

```
Invalid input
```

MLF

Section Id :	64065379123
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	13
Number of Questions to be attempted :	13
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653168514
Question Shuffling Allowed :	No

Question Number : 83 Question Id : 6406531113776 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533774204. YES